



NLP Assignment 3

Comparative Evaluation of Full Fine-Tuning and PEFT Methods for IMDb Sentiment Classification

Course: Natural Language Processing
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Abstract—This report evaluates Full Fine-Tuning, LoRA, QLoRA, and IA3 for IMDb binary sentiment classification. The comparison reports accuracy, F1-macro, training time, peak GPU memory, and trainable parameters, with results aggregated over seeds 42 and 43. RoBERTa-base Full Fine-Tuning with learning rate 2×10^{-5} gives the strongest valid result, achieving $93.85 \pm 0.04\%$ accuracy and $93.85 \pm 0.04\%$ F1-macro. The best PEFT result is RoBERTa-base LoRA with rank 16 and Attention+FFN placement at 2×10^{-5} , achieving $91.97 \pm 0.07\%$ accuracy. QLoRA provides the best memory-efficiency trade-off, reaching $89.54 \pm 0.13\%$ accuracy with 0.626 ± 0.006 GB peak VRAM. IA3 is the fastest selected valid configuration but remains less accurate than LoRA and QLoRA. DeBERTa-v3 results are shown as N/A where the available logs were incomplete or invalid for final comparison.

Index Terms—Full Fine-Tuning, LoRA, QLoRA, IA3, PEFT, IMDb, sentiment classification, Pareto analysis.

I. INTRODUCTION

Transformer models can be adapted to a downstream task either by updating the complete pretrained model or by training a smaller set of task-specific parameters. In this assignment, the downstream task is IMDb sentiment classification, and the comparison focuses on both predictive performance and computational efficiency.

Four fine-tuning techniques are used. **Full Fine-Tuning** updates all pretrained transformer weights together with the classification head, so it is expected to provide the strongest accuracy baseline but requires the largest trainable parameter budget. **LoRA** freezes the original model weights and learns small low-rank update matrices inside selected transformer layers, which reduces the number of trainable parameters while still adapting the model to the task. **QLoRA** further improves memory efficiency by keeping the base model quantized, using NF4 quantization in the logged experiments, while training LoRA parameters on top of the quantized model. **IA3** uses lightweight multiplicative adapter vectors in selected attention and feed-forward components, allowing only a very small set of adapter parameters to be trained.

The objective is to compare these methods using accuracy, F1-macro, training time, peak VRAM, and trainable parameters. The analysis then uses the assignment-required ablations and Pareto plots to identify which configurations provide the best trade-off between accuracy and efficiency.

II. EXPERIMENTAL SETUP

A. Dataset and Preprocessing

The task is binary sentiment classification using IMDb from Hugging Face. Each notebook loads the dataset, applies the appropriate tokenizer for the selected model, removes unused columns, renames the label column to `labels`, and uses padding/truncation. The implementation follows the design-matrix default of `max_length=256`, `epochs = 2`, `LR =  $1 \times 10^{-5}$` unless varied, and seeds 42 and 43.

B. Models and Methods

The intended model families are RoBERTa-base, DistilBERT, and DeBERTa-v3. The final valid comparison uses RoBERTa-base and DistilBERT results. DeBERTa-v3 is reported as N/A where the logs were incomplete, invalid for comparison, or not available for QLoRA. The methods are:

- **Full Fine-Tuning**: all model weights trainable.
- **LoRA**: rank $r = 8$ and $r = 16$ attention-only ablations, plus rank 16 Attention+FFN placement.
- **QLoRA**: quantized base model with LoRA rank 16 and NF4 configuration.
- **IA3**: attention-only and Attention+FFN adapter placements as recorded in the experiment logs.

C. Hardware, Training, and Logging

Experiments were executed in Kaggle notebooks using CUDA. The notebook logs show Tesla T4 GPUs with 15 GB memory each. Hugging Face Trainer was used for training and evaluation. Comparable runs used the same two seeds, two epochs, tokenizer policy, and batch/accumulation settings. Metrics were logged as accuracy, F1-macro, training time in seconds, peak VRAM in GB, and trainable parameters in millions. Trainable parameter counts were added from the notebook outputs and model-level counts because they were not stored in the CSV files.

III. RESULTS AND VISUALIZATIONS

Table III summarizes the best valid configuration for each method. The complete aggregated table is included in Appendix A.

TABLE I: File-to-experiment mapping

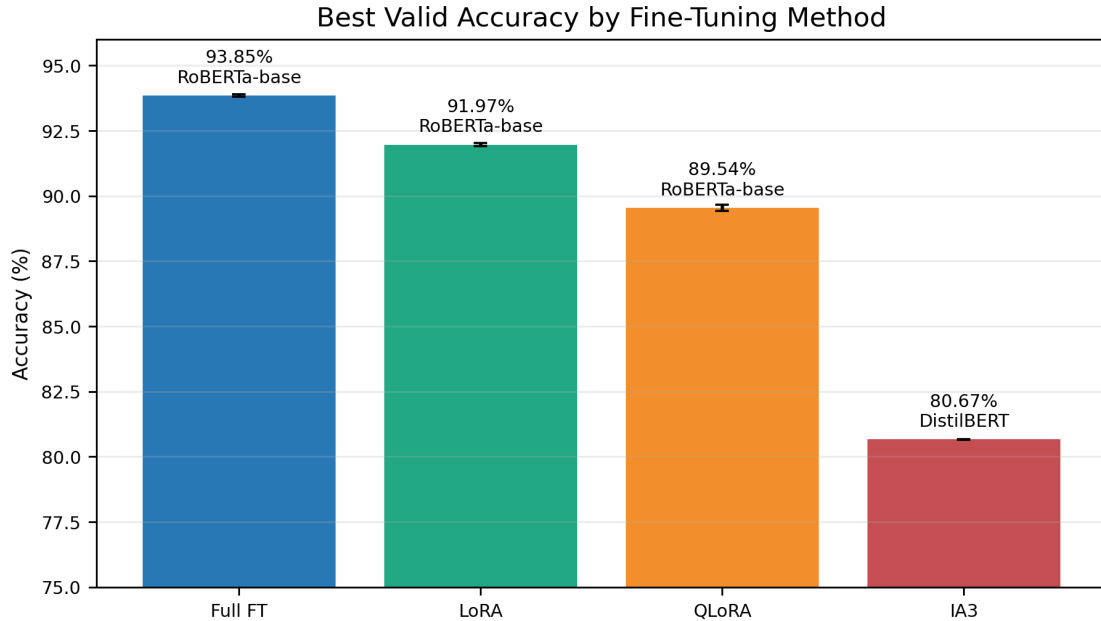
Part / ExpID	Requirement	File(s)	Status
Part 1	Data preprocessing	All notebooks	IMDb loaded from Hugging Face, tokenized with model tokenizer, max_length=256, padding/truncation applied.
ExpID 01	Full Fine-Tuning, LR 1e-5	1-full-fine-tuning.ipynb; full_ft_experiment_results.csv	1- RoBERTa-base and DistilBERT valid; DeBERTa-v3 reported N/A.
ExpID 02-04	LoRA rank and placement	2,3,4-lora_all_experiments.ipynb; lora_experiment_results.csv	2,3,4- r=8/r=16 attention-only and r=16 Attention+FFN covered for RoBERTa-base and DistilBERT; DeBERTa-v3 N/A.
ExpID 05	QLoRA NF4 r=16	5-qlora.ipynb; 5-qlora_experiment_results.csv	RoBERTa-base and DistilBERT valid; QLoRA DeBERTa-v3 not run.
ExpID 06	IA3 placement	6-ia3-adapter.ipynb; ia3_adapter_experiment_results.csv	Attention-only and Attention+FFN configurations logged; DeBERTa-v3 N/A.
ExpID 07	LR sensitivity, Full FT	exp07_full-fine-tuning_2e-5lr.ipynb; exp07_full_ft_experiment_results.csv	LR changed from 1e-5 to 2e-5 for Full Fine-Tuning.
ExpID 07	LR sensitivity, PEFT	exp07-lr-2e-5_lora_qlora_adapter.ipynb; exp07_lora/qlora/ia3 CSVs	Best available PEFT configurations rerun at LR=2e-5.
ExpID 08	Low-resource data fraction	Not included	Optional experiment skipped; no 30% data robustness result is reported.

TABLE II: Requirement coverage summary

Required item	Status	Where addressed
IMDb loading, tokenization, padding/truncation	Covered	Preprocessing is implemented in the notebooks for each selected model.
Full Fine-Tuning, LoRA, QLoRA, IA3	Covered	ExpID 01, 02-06, and 07 tables; method comparison figures.
Seeds 42 and 43 with mean $\pm$ standard deviation	Covered	Main tables and Appendix A.
Accuracy and F1-macro	Covered	Tables III and Appendix A; Figs. 1, 4, and 6.
Training time, peak VRAM, trainable parameters	Covered	Tables III and Appendix A; Figs. 2, 3, and 7.
LoRA rank sensitivity	Covered	Fig. 4 and Section IV-B.
Pareto curves	Covered	VRAM-vs-accuracy and time-vs-accuracy in Figs. 2 and 3.
Module placement analysis	Covered	Fig. 5 and Section IV-C.
LR 1×10^{-5} to 2×10^{-5} sensitivity	Covered	Fig. 6 and Section IV-E.
ExpID 08 low-resource run	Optional; not included	Stated in file mapping, analysis, and conclusion.

TABLE III: Best valid configuration per method. Values are mean $\pm$ standard deviation over seeds 42 and 43.

Method	Model	Configuration	LR	Accuracy (%)	F1 (%)	Time (s)	VRAM (GB)	Params (M)
Full FT	RoBERTa-base	Full Fine-Tuning	2e-5	93.85 $\pm$ 0.04	93.85 $\pm$ 0.04	468.8 $\pm$ 9.5	3.678 $\pm$ 0.000	124,647
LoRA	RoBERTa-base	r=16, Attn+FFN	2e-5	91.97 $\pm$ 0.07	91.96 $\pm$ 0.07	546.7 $\pm$ 18.1	2.644 $\pm$ 0.000	2.951
QLoRA	RoBERTa-base	NF4, r=16, double quantization on	2e-5	89.54 $\pm$ 0.13	89.54 $\pm$ 0.13	715.6 $\pm$ 4.5	0.626 $\pm$ 0.006	1.182
IA3	DistilBERT	Attn+FFN	2e-5	80.67 $\pm$ 0.01	80.67 $\pm$ 0.01	221.4 $\pm$ 0.9	2.644 $\pm$ 0.000	0.620

**Fig. 1:** Best valid accuracy per fine-tuning method. Error bars show standard deviation over seeds 42 and 43.

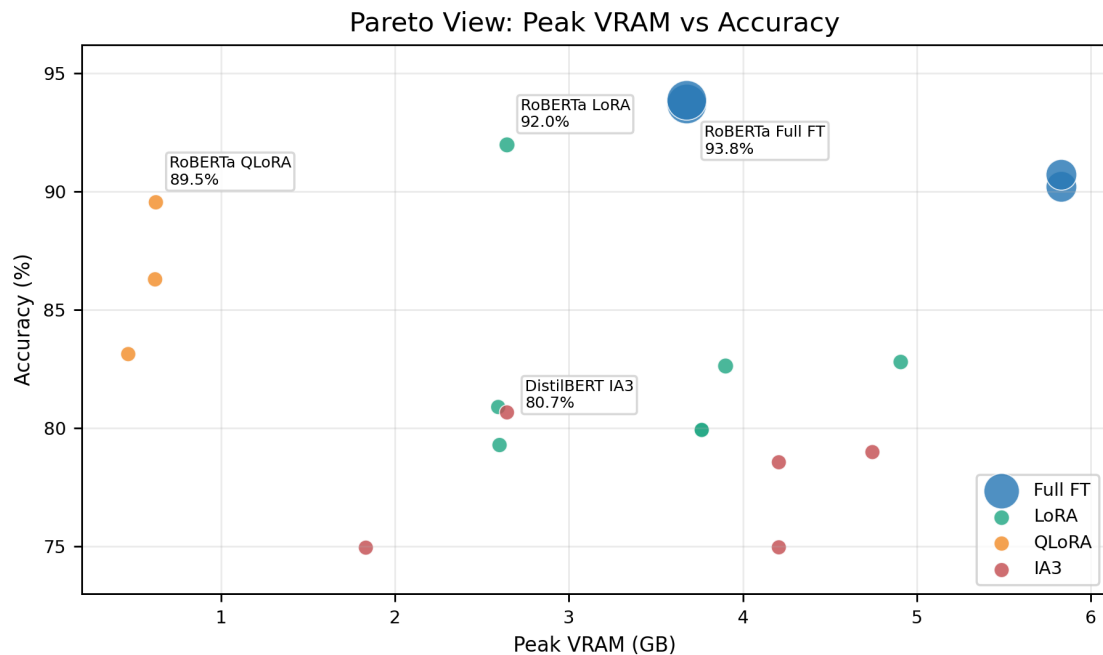


Fig. 2: Pareto view for peak VRAM versus accuracy. QLoRA gives the strongest memory-efficiency point, while Full Fine-Tuning gives the strongest accuracy point.

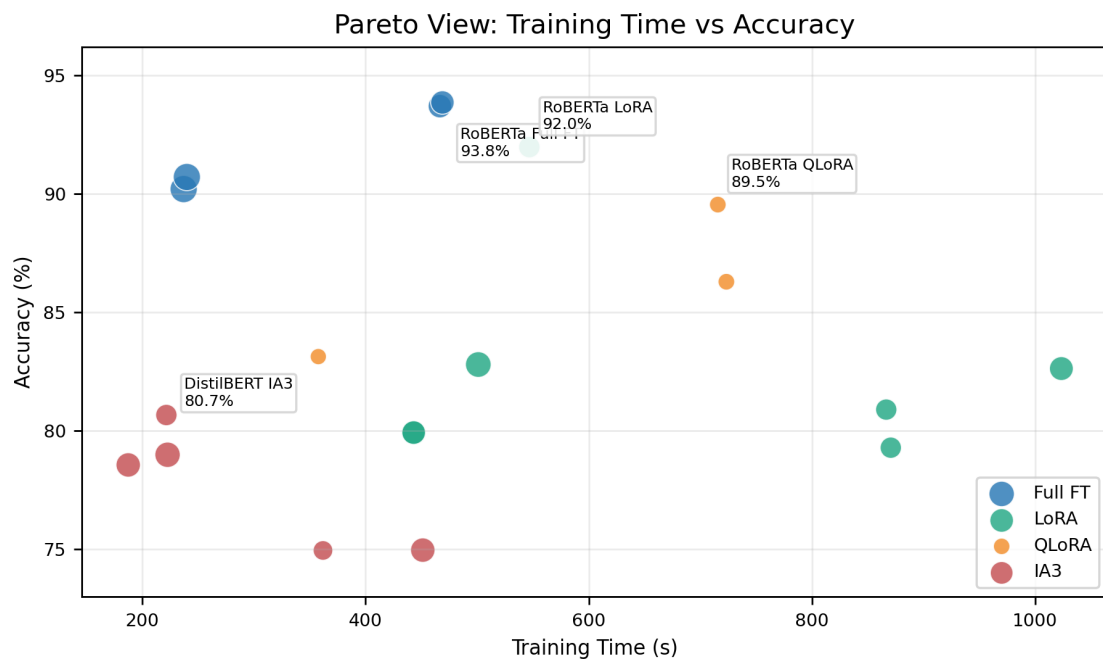


Fig. 3: Pareto view for training time versus accuracy. Bubble size represents peak VRAM.

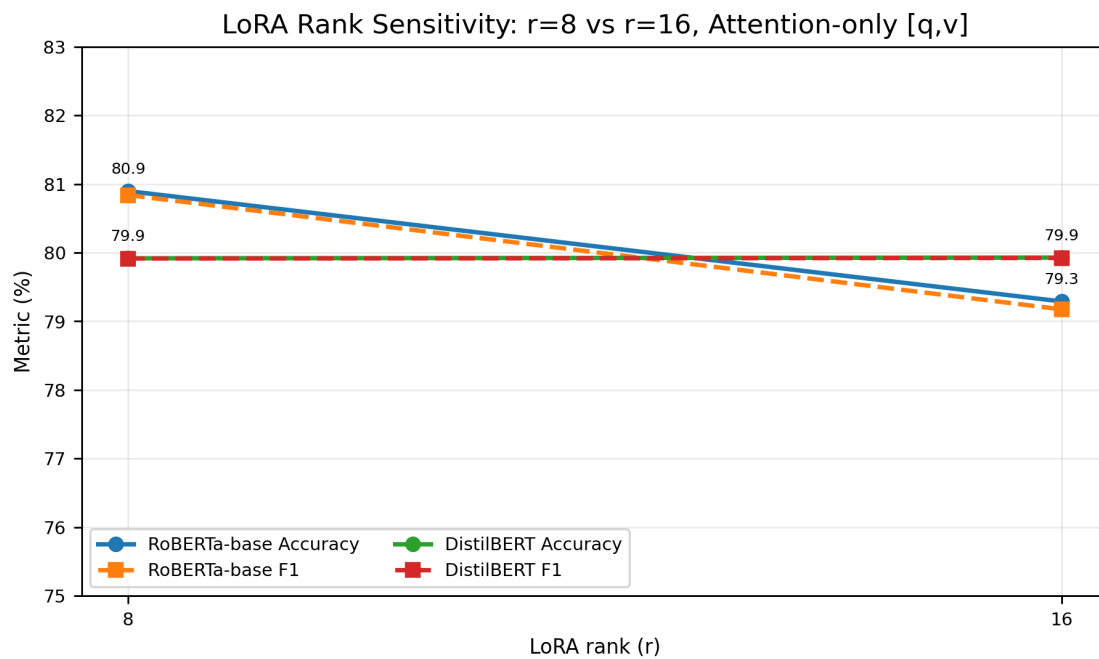


Fig. 4: LoRA rank sensitivity for $r = 8$ and $r = 16$ under attention-only [q,v] settings.

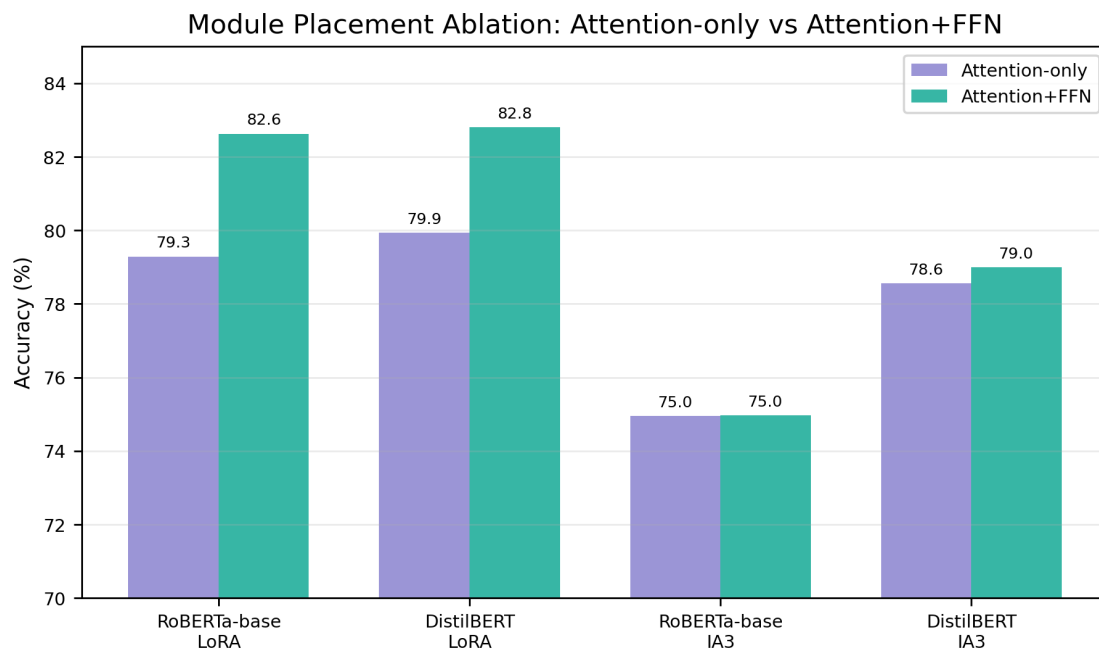
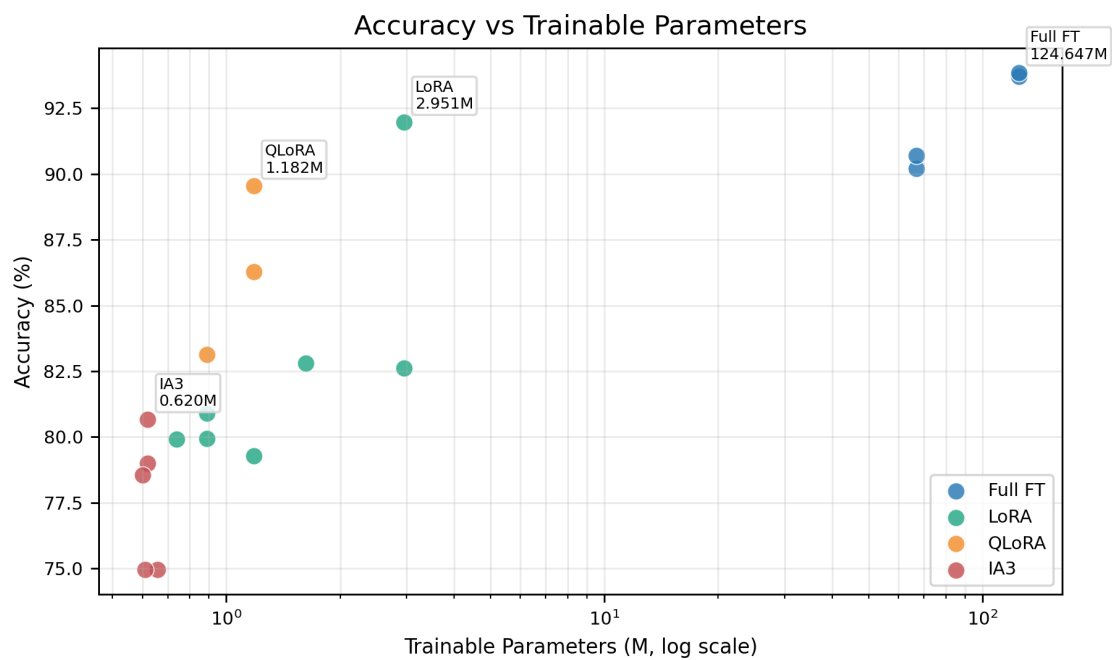
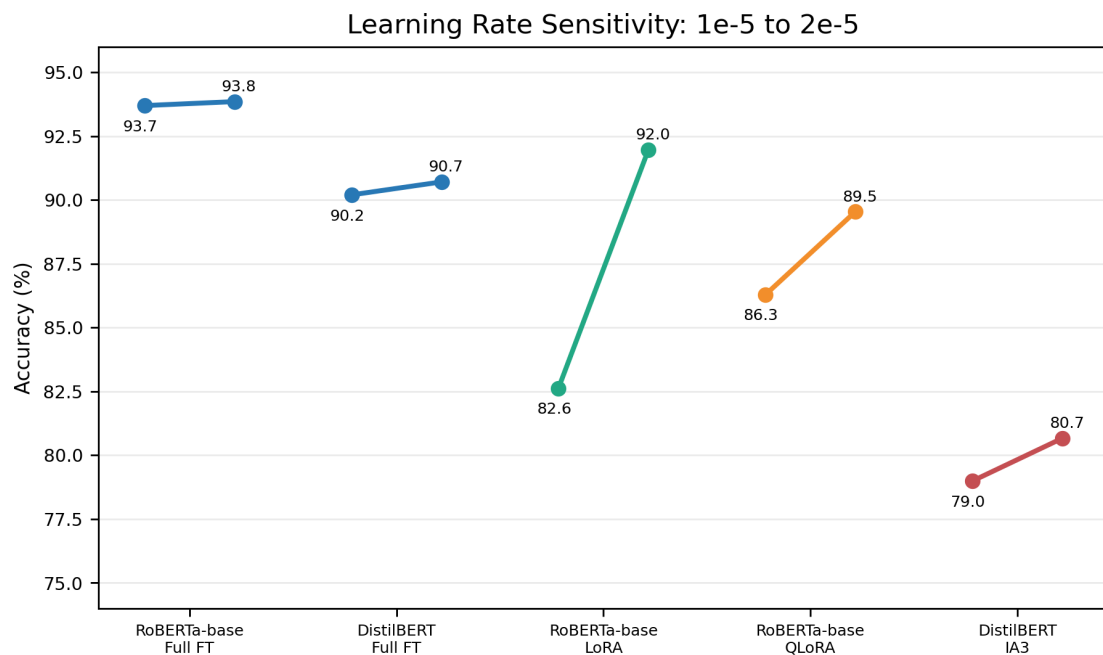


Fig. 5: Module placement ablation for Attention-only versus Attention+FFN settings in LoRA and IA3.



IV. ANALYSIS AND DISCUSSION

A. Pareto Frontier

RoBERTa-base Full Fine-Tuning at $LR\ 2 \times 10^{-5}$ is the strongest accuracy point, with $93.85 \pm 0.04\%$ accuracy, 3.678 ± 0.000 GB peak VRAM, and 124.647M trainable parameters. RoBERTa-base LoRA at $LR\ 2 \times 10^{-5}$ is the strongest PEFT accuracy point, with $91.97 \pm 0.07\%$ accuracy and only 2.951M trainable parameters. QLoRA is the strongest memory-efficient point, reaching $89.54 \pm 0.13\%$ accuracy with 0.626 ± 0.006 GB peak VRAM.

B. LoRA Rank Sensitivity

Increasing rank from $r = 8$ to $r = 16$ did not improve attention-only LoRA in these runs. RoBERTa-base decreased from $80.90 \pm 2.37\%$ to $79.29 \pm 0.38\%$, while DistilBERT stayed almost unchanged at about 79.9%. Under this setup, rank alone was less important than module placement and learning rate.

C. Module Placement

LoRA benefited from Attention+FFN placement. For RoBERTa-base at $LR\ 1 \times 10^{-5}$, $r=16$ attention-only reached

79.29%, while $r=16$ Attention+FFN reached 82.63%. For DistilBERT, the corresponding improvement was from 79.93% to 82.80%. IA3 showed a smaller placement effect: RoBERTa-base stayed near 75%, and DistilBERT improved from 78.56% to 78.99% at $LR\ 1 \times 10^{-5}$.

D. QLoRA Efficiency

QLoRA reduced VRAM substantially. Compared with RoBERTa-base Full Fine-Tuning at $LR\ 2 \times 10^{-5}$, RoBERTa-base QLoRA at $LR\ 2 \times 10^{-5}$ used about 3.05 GB less logged peak VRAM and required only 1.182M trainable parameters instead of 124.647M. The accuracy trade-off was approximately 4.31 percentage points.

E. Optimization Sensitivity

Changing the learning rate from 1×10^{-5} to 2×10^{-5} improved the best available PEFT runs. RoBERTa-base LoRA Attention+FFN improved from 82.63% to 91.97%, RoBERTa-base QLoRA improved from 86.29% to 89.54%, and DistilBERT IA3 Attention+FFN improved from 78.99% to 80.67%. Full Fine-Tuning improved only slightly for RoBERTa-base and DistilBERT.

F. Robustness

ExpID 08, the 30% low-resource experiment, was optional and was not included. Therefore, no claim is made about low-resource robustness. The report focuses on the required ExpID 01-07 comparisons and the available LR sensitivity runs.

V. CONCLUSION

Full Fine-Tuning remains the best accuracy anchor, with RoBERTa-base reaching 93.85% accuracy at $LR\ 2 \times 10^{-5}$. LoRA is the best PEFT method in terms of accuracy, especially when Attention+FFN placement and $LR\ 2 \times 10^{-5}$ are used. QLoRA provides the best VRAM/accuracy trade-off and is the strongest choice under memory constraints. IA3 is lightweight and fast on the selected DistilBERT configuration, but it is less accurate than LoRA and QLoRA in these experiments. DeBERTa-v3 is not included in the final comparison and is reported as N/A.

APPENDIX A

COMPLETE AGGREGATED RESULTS

All values are mean $\pm$ sample standard deviation over seeds 42 and 43. DeBERTa-v3 entries are N/A.

TABLE IV: Complete result summary

Exp	Method	Model	Config	LR	Acc. (%)	F1 (%)	Time (s)	VRAM (GB)	Params (M)
01	Full FT	DeBERTa-v3	Full Fine-Tuning	1e-5	N/A	N/A	N/A	N/A	N/A
01	Full FT	DistilBERT	Full Fine-Tuning	1e-5	90.20 $\pm$ 0.01	90.19 $\pm$ 0.01	237.0 $\pm$ 3.9	5.831 $\pm$ 0.000	66.955
01	Full FT	RoBERTa-base	Full Fine-Tuning	1e-5	93.69 $\pm$ 0.24	93.69 $\pm$ 0.24	466.5 $\pm$ 18.1	3.678 $\pm$ 0.000	124.647
02	LoRA	DeBERTa-v3	r=8, Attn-only [q,v]	1e-5	N/A	N/A	N/A	N/A	N/A
02	LoRA	DistilBERT	r=8, Attn-only [q,v]	1e-5	79.92 $\pm$ 0.11	79.91 $\pm$ 0.11	443.3 $\pm$ 0.4	3.764 $\pm$ 0.000	0.740
02	LoRA	RoBERTa-base	r=8, Attn-only [q,v]	1e-5	80.90 $\pm$ 2.37	80.84 $\pm$ 2.44	866.6 $\pm$ 2.2	2.594 $\pm$ 0.001	0.887
03	LoRA	DeBERTa-v3	r=16, Attn-only [q,v]	1e-5	N/A	N/A	N/A	N/A	N/A
03	LoRA	DistilBERT	r=16, Attn-only [q,v]	1e-5	79.93 $\pm$ 0.01	79.92 $\pm$ 0.01	443.1 $\pm$ 0.3	3.764 $\pm$ 0.000	0.887
03	LoRA	RoBERTa-base	r=16, Attn-only [q,v]	1e-5	79.29 $\pm$ 0.38	79.17 $\pm$ 0.40	870.6 $\pm$ 0.0	2.602 $\pm$ 0.000	1.182
04	LoRA	DeBERTa-v3	r=16, Attn+FFN	1e-5	N/A	N/A	N/A	N/A	N/A
04	LoRA	DistilBERT	r=16, Attn+FFN	1e-5	82.80 $\pm$ 0.09	82.79 $\pm$ 0.09	501.0 $\pm$ 0.2	4.907 $\pm$ 0.000	1.624
04	LoRA	RoBERTa-base	r=16, Attn+FFN	1e-5	82.63 $\pm$ 0.58	82.43 $\pm$ 0.63	1023.5 $\pm$ 1.6	3.901 $\pm$ 0.000	2.951
05	QLoRA	DeBERTa-v3	NF4, r=16, double quantization on	1e-5	N/A	N/A	N/A	N/A	N/A
05	QLoRA	DistilBERT	NF4, r=16, double quantization on	1e-5	83.13 $\pm$ 0.03	83.13 $\pm$ 0.03	357.6 $\pm$ 3.1	0.467 $\pm$ 0.001	0.887
05	QLoRA	RoBERTa-base	NF4, r=16, double quantization on	1e-5	86.29 $\pm$ 1.18	86.28 $\pm$ 1.18	723.3 $\pm$ 2.3	0.622 $\pm$ 0.000	1.182
06	IA3	DeBERTa-v3	Attn+FFN	1e-5	N/A	N/A	N/A	N/A	N/A
06	IA3	DeBERTa-v3	Attn-only	1e-5	N/A	N/A	N/A	N/A	N/A
06	IA3	DistilBERT	Attn+FFN	1e-5	78.99 $\pm$ 0.00	78.99 $\pm$ 0.00	222.5 $\pm$ 0.9	4.745 $\pm$ 0.000	0.620
06	IA3	DistilBERT	Attn-only	1e-5	78.56 $\pm$ 0.00	78.56 $\pm$ 0.00	187.3 $\pm$ 0.2	4.207 $\pm$ 0.000	0.601
06	IA3	RoBERTa-base	Attn+FFN	1e-5	74.97 $\pm$ 0.03	74.32 $\pm$ 0.03	451.2 $\pm$ 21.2	4.207 $\pm$ 0.000	0.657
06	IA3	RoBERTa-base	Attn-only	1e-5	74.96 $\pm$ 0.05	74.32 $\pm$ 0.05	361.8 $\pm$ 14.0	1.833 $\pm$ 0.000	0.611
07	Full FT	DeBERTa-v3	Full Fine-Tuning	2e-5	N/A	N/A	N/A	N/A	N/A
07	Full FT	DistilBERT	Full Fine-Tuning	2e-5	90.70 $\pm$ 0.02	90.70 $\pm$ 0.02	239.8 $\pm$ 4.0	5.831 $\pm$ 0.000	66.955
07	Full FT	RoBERTa-base	Full Fine-Tuning	2e-5	93.85 $\pm$ 0.04	93.85 $\pm$ 0.04	468.8 $\pm$ 9.5	3.678 $\pm$ 0.000	124.647
07	IA3	DistilBERT	Attn+FFN	2e-5	80.67 $\pm$ 0.01	80.67 $\pm$ 0.01	221.4 $\pm$ 0.9	2.644 $\pm$ 0.000	0.620
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07	QLoRA	RoBERTa-base	NF4, r=16, double quantization on	2e-5	89.54 $\pm$ 0.13	89.54 $\pm$ 0.13	715.6 $\pm$ 4.5	0.626 $\pm$ 0.006	1.182